# Retail Sales Data Cleaning and Analysis

**Prepared by:** Okewu Jeremiah
**Tools:** Python, pandas, Matplotlib and Excel
**Dataset:** 50,000 synthetic retail transactions

## Executive Summary

The analysis covered 50,000 transaction records with a combined transaction value of NGN 4,998,963,652.92. The average transaction value was NGN 99,979.27. The work included data inspection, date conversion, validity checks, outlier screening, and sales analysis by month, store, city and payment method.

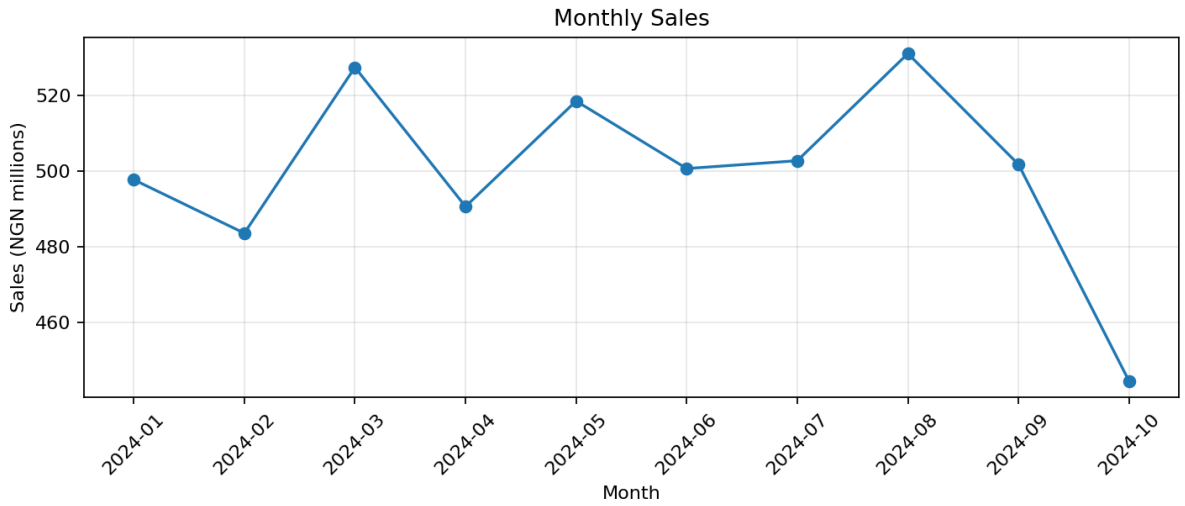
| Metric | Result |
|---|---|
| Transaction records | 50,000 |
| Total transaction value | NGN 4,998,963,652.92 |
| Average transaction value | NGN 99,979.27 |
| Missing values detected initially | 0 |
| Exact duplicate rows detected initially | 0 |

## 1. Data Quality and Preparation

The initial inspection found no missing values or fully duplicated rows. The transaction date column was converted from text to datetime. No invalid dates, non-positive transaction amounts, or non-positive item counts were detected. The IQR method did not flag outliers in the three numeric columns tested. These checks do not prove that every value is correct.

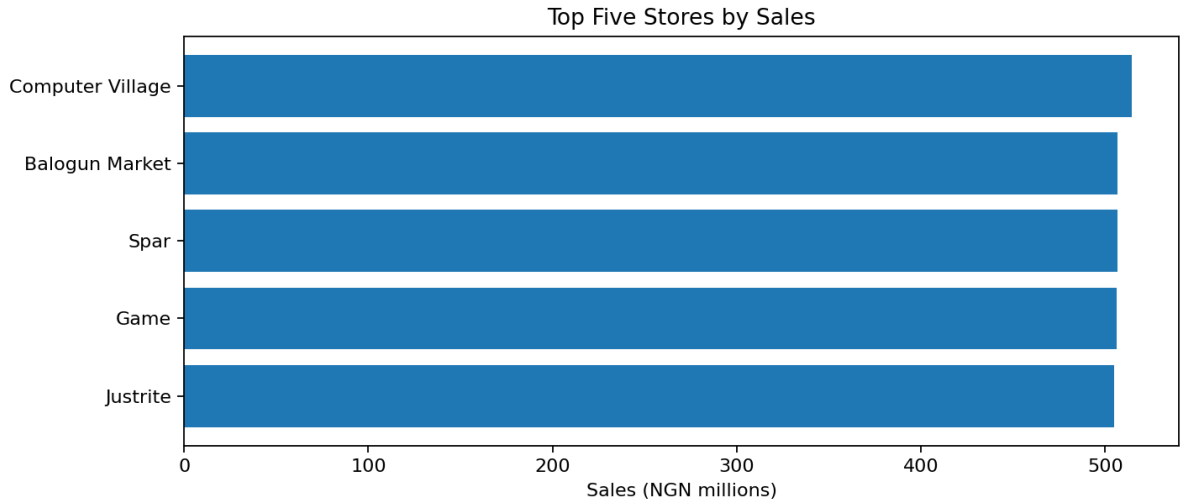
## 2. Monthly Sales Analysis

2024-08 had the highest monthly sales, at NGN 531,090,142.35. 2024-10 had the lowest, at NGN 444,534,909.26. Monthly summaries help compare transaction activity over time.




## 3. Store Performance

Computer Village ranked first by total sales at NGN 514,221,321.08. Mr Price ranked last at NGN 486,171,392.63. These rankings identify differences for further investigation, but do not explain why the differences occurred.

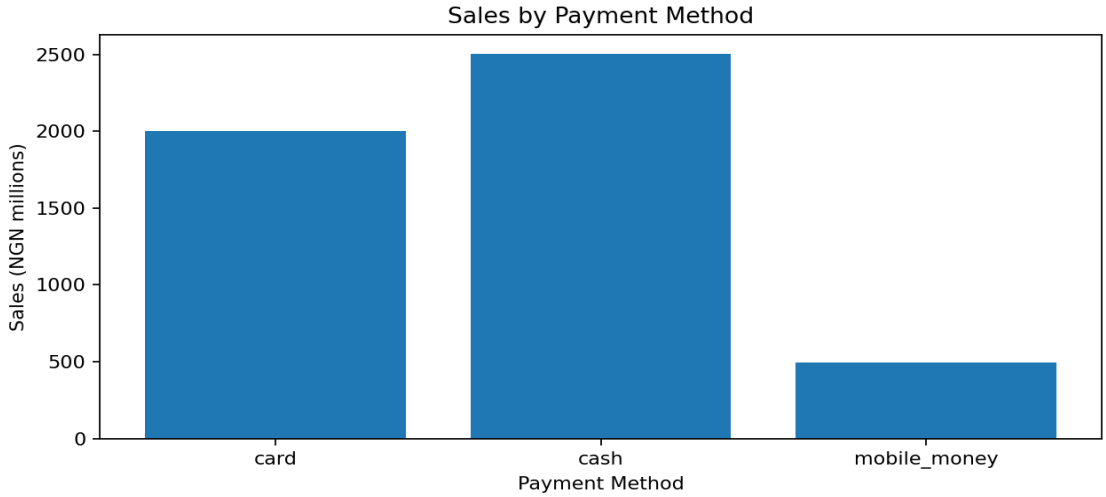



## 4. City Performance

Calabar recorded the highest city-level sales at NGN 348,517,159.27, while Aba recorded the lowest at NGN 313,506,505.25. Further analysis would be required to understand the causes of these differences.

## 5. Payment Method Analysis

Cash was the most frequent payment method, with 25,084 transactions. Card transactions had an average transaction value of NGN 100,559.20. Transaction frequency and average transaction value measure different aspects of payment behaviour.




## 6. Recommendations

• Monitor monthly sales and transaction counts to identify changes in activity.

• Investigate store and city differences before making operational decisions.

• Track payment method usage to understand transaction preferences.

• Validate future datasets against business rules and investigate suspicious records.

• For a real company engagement, repeat the analysis using authorized, verified company data.

## 7. Scope and Limitations

The dataset is synthetic and covers transactions from January 1 to October 27, 2024. It does not cover the full 2018–2026 period and must not be interpreted as verified sales from actual businesses. Store and city names in the dataset do not establish real-world performance. No records were removed based on the checks performed in this project.

Prepared as a portfolio demonstration by Okewu Jeremiah.